

Electronics Homework 4 : Common-Gate Current Buffer Design

Chao-Yang Zhan

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Design Specifications:

- $V_{DD} = 1.8\text{V}$, $R_4 = 100\text{k}\Omega$.
- Transistor: $V_{th} = 0.35\text{V}$, $k_p = 200\mu\text{A}/\text{V}^2$.
- Targets: Current Gain > 0.95 A/A, Total Power $< 0.6\text{mW}$.

Problem 1 (*Common-Gate Amplifier Design*)

(a) Power Consumption and Bias Current Design

Given $P_{total} < 0.6\text{mW}$ and $V_{DD} = 1.8\text{V}$, the total current budget is:

$$I_{total} < \frac{0.6\text{mW}}{1.8\text{V}} \approx 333\mu\text{A}.$$

We select a drain current $I_D = 9\mu\text{A}$ and a bias network current $I_{bias} \approx 1.8\mu\text{A}$.

$$I_{total} = I_D + I_{bias} = 10.8\mu\text{A} \implies P_{total} = 1.8\text{V} \times 10.8\mu\text{A} \approx 19.44\mu\text{W} \ll 0.6\text{mW}.$$

(b) DC Node Voltages and Component Sizing

1. Source Resistor (R_3): Setting $I_D = 9\mu\text{A}$ and $R_3 = 50\text{k}\Omega$:

$$V_S = I_D \times R_3 = 9\mu\text{A} \times 50\text{k}\Omega = 0.45\text{V}.$$

2. Transistor M1 (W/L) Design: The current gain of the Common-Gate stage is determined by the ratio of the source resistance R_3 to the input impedance of the transistor $R_{in} = 1/g_m$:

$$A_i \approx \frac{R_3}{R_3 + R_{in}} = \frac{R_3}{R_3 + \frac{1}{g_m}}$$

To meet the requirement $A_i > 0.95$, we set a target gain of 0.967 and $R_3 = 50\text{k}\Omega$:

$$0.967 \approx \frac{50\text{k}}{50\text{k} + \frac{1}{g_m}} \implies g_m \approx 0.6\text{mS}$$

Using the transconductance formula $g_m = \sqrt{2k_p(W/L)I_D}$ with $I_D = 9\mu\text{A}$ and $k_p = 200\mu\text{A}/\text{V}^2$:

$$0.6\text{m} = \sqrt{2 \cdot 200\mu \cdot (W/L) \cdot 9\mu} \implies W/L \approx 100.$$

Thus, we choose $W = 50\mu\text{m}$ and $L = 0.5\mu\text{m}$ to satisfy the gain requirement.

3. Biasing (R_1, R_2) and DC Operating Point: To ensure the transistor M_1 operates in the saturation region, we must satisfy $V_{GS} > V_{th}$ and $V_D > V_G - V_{th}$.

First, with $V_S = 0.45\text{V}$ and $V_{th} = 0.35\text{V}$, we select $V_G = 0.9\text{V}$ to provide sufficient overdrive voltage:

$$V_{GS} = V_G - V_S = 0.9\text{V} - 0.45\text{V} = 0.45\text{V} > 0.35\text{V} = V_{th}$$

Second, to verify saturation, we check the condition $V_D > V_G - V_{th}$:

$$V_D = V_{DD} - I_D \cdot R_{load} = 1.8\text{V} - (9\mu\text{A} \cdot 100\text{k}\Omega) = 0.9\text{V}$$

$$V_D = 0.9\text{V} > 0.55\text{V} = 0.9\text{V} - 0.35\text{V} = V_G - V_{th}$$

Since $0.9\text{V} > 0.55\text{V}$, M_1 is confirmed to be in the saturation region. We set the biasing network current I_{bias} to $1.8\mu\text{A}$ (So $R_1 + R_2 = \frac{V_{DD}}{I_{bias}} = 1\text{M}\Omega$), and apply the voltage divider rule to obtain $V_G = 0.9\text{V}$:

$$V_G = V_{DD} \frac{R_2}{R_1 + R_2} \implies 0.9\text{V} = 1.8\text{V} \cdot \frac{R_2}{1\text{M}\Omega} \implies R_1 = 500\text{k}\Omega, R_2 = 500\text{k}\Omega.$$

(c) Schematic Design

The common-gate amplifier is implemented in SPICE with the parameters summarized in the following table. Figure 1 shows the final circuit schematic used for simulation.

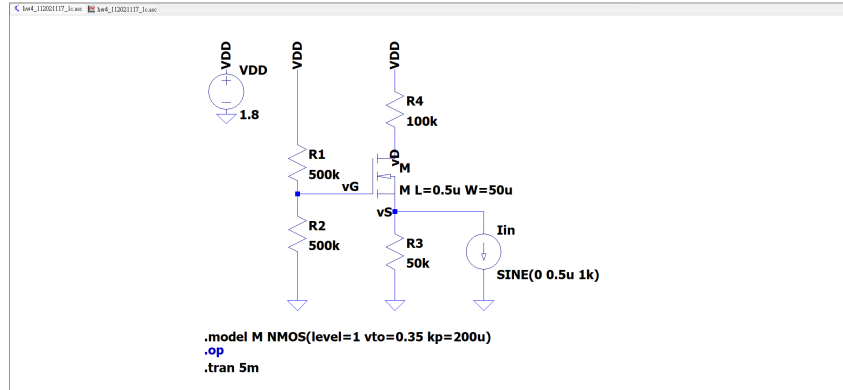


Figure 1: Common-Gate Amplifier SPICE Schematic

Component	Value
R_1, R_2	$500\text{k}\Omega$
R_3	$50\text{k}\Omega$
R_4	$100\text{k}\Omega$
M (W/L)	$50\mu\text{m}/0.5\mu\text{m}$

Table 1: Design Parameters

(d) Simulation Results and Verification

1. Power Consumption Analysis

Based on the operating point (.op) analysis, the total power consumption is verified:

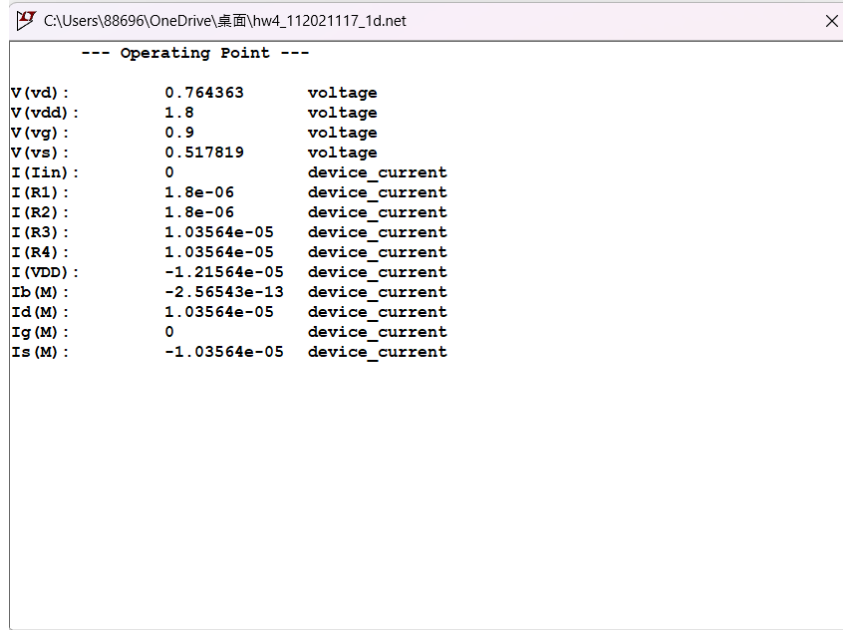


Figure 2: DC Operating Point Analysis

From the simulated operating point (.op), the total current drawn from V_{DD} consists of the bias network current ($I_{R1} = 1.8\mu A$) and the drain current ($I_D \approx 10.36\mu A$):

$$I_{total} = 1.8\mu A + 10.36\mu A = 12.16\mu A$$

The total power consumption is calculated as:

$$P_{total} = V_{DD} \cdot I_{total} = 1.8V \cdot 12.16\mu A \approx 21.88\mu W$$

This result is significantly lower than the specified 0.6mW limit, confirming the design is power-efficient.

2. Current Gain Verification

To observe the signal component, the DC offset of the load current $I(R_4)$ is removed by subtracting the quiescent current ($\approx 10.36\mu A$). Figure 3 shows the original output, while Figure 4 compares the input I_{in} and the processed output current I_{out} .

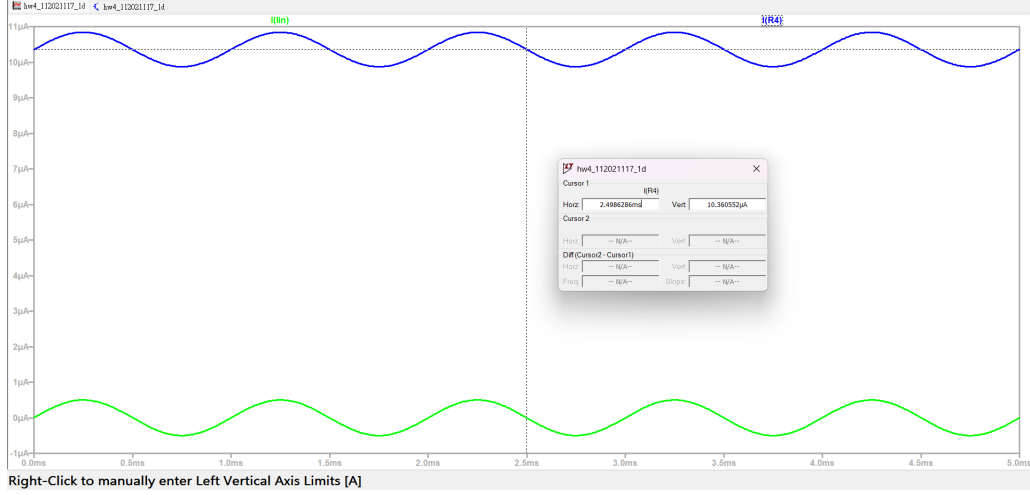
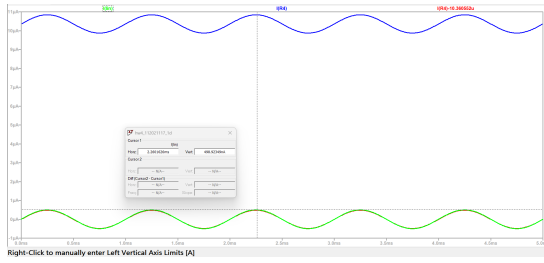
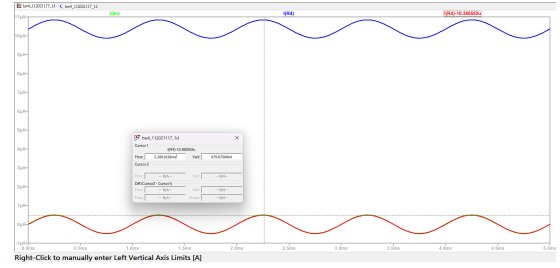


Figure 3: Original $I(R_4)$ with DC Offset



(a) Input $I_{in}(AC)$



(b) Output $I_{out}(AC)$

Figure 4: Output signal extraction and Gain verification

3. Performance Calculation

From Figure 4, the input signal amplitude is approximately 498.9nA, and the processed output signal amplitude is approximately 479.9nA. The current gain A_i is calculated as:

$$A_i = \frac{I_{out_AC}}{I_{in}} \approx \frac{479.9\text{nA}}{498.9\text{nA}} \approx 0.962 \text{ A/A}$$

Since $0.962 > 0.95$, the design successfully meets the performance requirements. The slight deviation from unity is due to the finite output resistance of the NMOS transistor.

Problem 2 (Cascode Current Buffer Design)

(a) Power Consumption and Bias Current Design

Given $P_{total} < 0.6\text{mW}$ and $V_{DD} = 1.8\text{V}$, we select a drain current $I_D = 20\mu\text{A}$ and a bias network current $I_{bias} \approx 2\mu\text{A}$ to ensure robust operation:

$$I_{total} = I_D + I_{bias} = 22\mu\text{A} \implies P_{total} = 1.8\text{V} \times 22\mu\text{A} = 39.6\mu\text{W} \ll 0.6\text{mW}.$$

(b) DC Node Voltages and Component Sizing

1. Biasing and Operating Point Analysis:

We set the bias voltage $V_G = 0.9\text{V}$ using a voltage divider ($R_1 = R_2 = 500\text{k}\Omega$). Given

M_2 is diode-connected, its gate-source voltage V_{GS2} defines the source node voltage V_S . To maintain $I_D = 20\mu\text{A}$ at $V_S \approx 0.5\text{V}$, we solve for W_2 using the saturation current equation:

$$I_D = \frac{1}{2}k_p \left(\frac{W_2}{L} \right) (V_{GS2} - V_{th})^2$$

With $V_{GS2} = V_S \approx 0.5\text{V}$, $V_{th} = 0.35\text{V}$, and $k_p = 200\mu\text{A}/\text{V}^2$:

$$20\mu\text{A} = \frac{1}{2}(200\mu\text{A}/\text{V}^2) \left(\frac{W_2}{0.5\mu\text{m}} \right) (0.5\text{V} - 0.35\text{V})^2 \implies W_2 \approx 1.11\mu\text{m}$$

We select $W_2 = 1\mu\text{m}$ as the practical design value. We verify M_1 saturation:

$$V_{GS1} = V_G - V_S = 0.9\text{V} - 0.5\text{V} = 0.4\text{V} > V_{th} = 0.35\text{V}$$

$$V_{DS1} = (V_{DD} - I_D R_4) - V_S = (1.8\text{V} - 0.4\text{V}) - 0.5\text{V} = 0.9\text{V} > (V_{GS1} - V_{th}) = 0.05\text{V}$$

2. Transistor Sizing (W/L) Optimization:

With the DC operating point secured, we optimize the current gain $A_i > 0.95$. The current I_{in} splits between the source of M_1 and the diode-connected M_2 . The gain is given by:

$$A_i \approx \frac{1}{1 + g_{m2}/g_{m1}} > 0.95 \implies \frac{g_{m1}}{g_{m2}} > 19$$

Using the square-law relationship $g_m \propto \sqrt{W}$, we require $\sqrt{W_1/W_2} > 19 \implies W_1/W_2 > 361$. Given $W_2 = 1\mu\text{m}$, we select $W_1 = 380\mu\text{m}$ ($L = 0.5\mu\text{m}$ for both), achieving:

$$\frac{g_{m1}}{g_{m2}} = \sqrt{380} \approx 19.5 \implies A_i \approx 0.951$$

(c) Schematic Design

The Cascode current buffer is implemented in SPICE with the optimized parameters summarized in the following table. Figure 5 shows the final circuit schematic used for simulation.

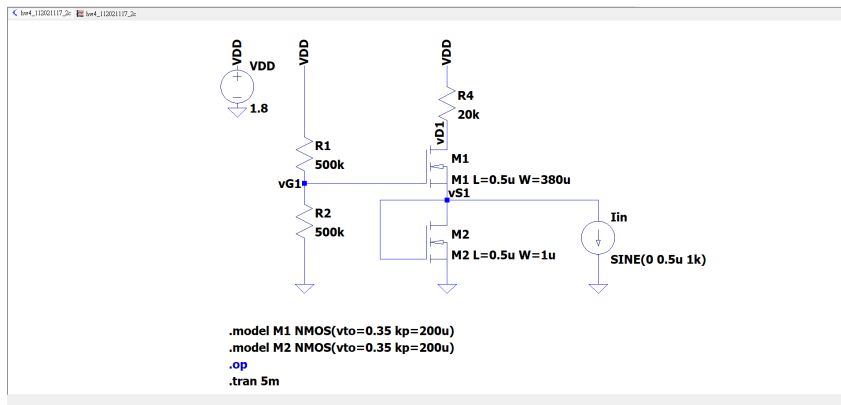


Figure 5: Cascode Current Buffer SPICE Schematic

Component	Value
R_1, R_2	500k Ω
R_4	20k Ω
M_1 (W/L)	380 $\mu\text{m}/0.5\mu\text{m}$
M_2 (W/L)	1 $\mu\text{m}/0.5\mu\text{m}$

Table 2: Design Parameters for Cascode Configuration

(d) Simulation Results and Verification

1. Power Consumption Analysis

Based on the operating point (.op) analysis, the total power consumption is verified:

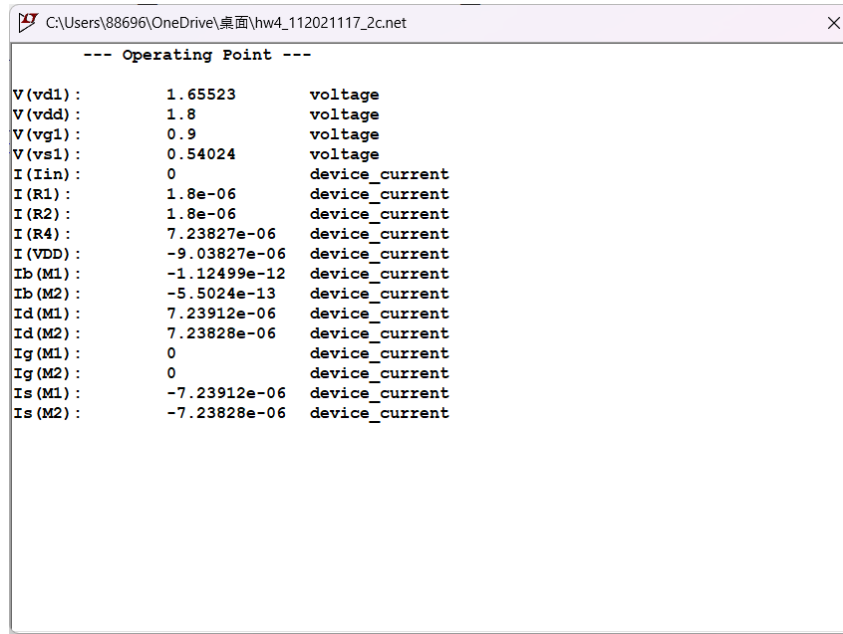


Figure 6: DC Operating Point Analysis

From the simulated operating point, the total current drawn from V_{DD} is the sum of the bias network current ($I_{R1} = 1.8\mu\text{A}$) and the drain current ($I_D \approx 7.24\mu\text{A}$):

$$I_{total} = 1.8\mu\text{A} + 7.24\mu\text{A} = 9.04\mu\text{A}$$

The total power consumption is:

$$P_{total} = V_{DD} \cdot I_{total} = 1.8\text{V} \cdot 9.04\mu\text{A} \approx 16.27\mu\text{W} \ll 0.6\text{mW}$$

2. Current Gain Verification

Figure 3 shows the original output current $I(R_4)$, and Figure 4 compares the input current I_{in} and the output current I_{out} .

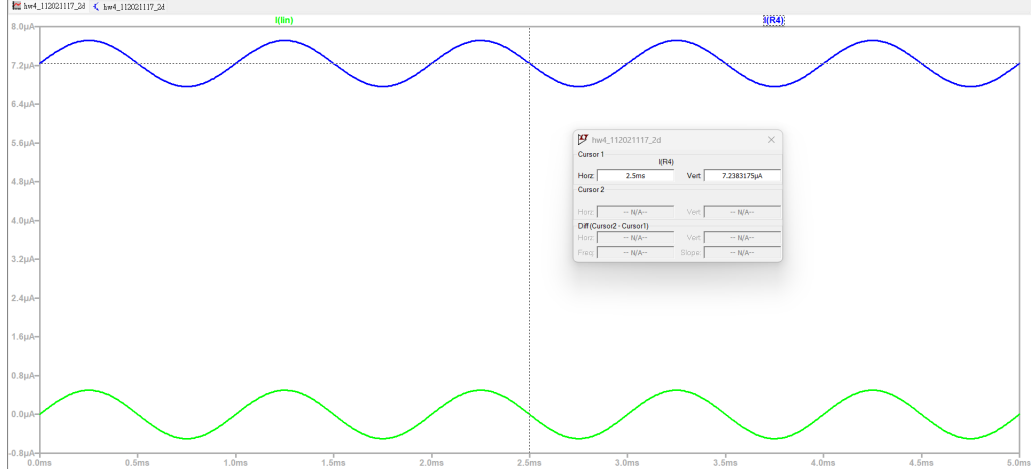
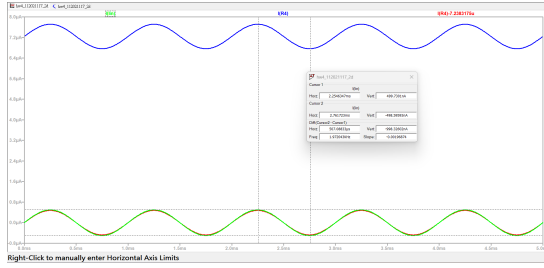
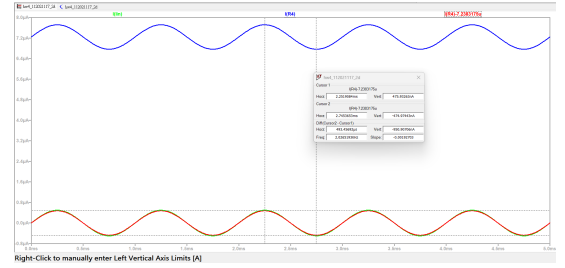


Figure 7: Original $I(R_4)$ with DC Offset



(a) Input $I_{in}(AC)$



(b) Output $I_{out}(AC)$

Figure 8: Output signal extraction and Gain verification

3. Performance Calculation

From the simulation, the input signal amplitude is 499.7nA and the output signal amplitude is 475.9nA. The current gain A_i is:

$$A_i = \frac{I_{out_AC}}{I_{in}} \approx \frac{475.9nA}{499.7nA} \approx 0.952 \text{ A/A}$$

The design meets the $A_i > 0.95$ requirement.

4. Summary and Comparison

Parameter	Problem 1 (CG)	Problem 2 (Cascode)
Current Gain (A_i)	0.962	0.952
Input Impedance	Low ($1/g_{m1}$)	Optimized ($1/g_{m1} \parallel 1/g_{m2}$)
Design Flexibility	Limited	High (Ratio-based)

Table 3: Comparison of CG and Cascode architectures

The Cascode architecture offers distinct advantages over the standard CG amplifier:

- **Precise Gain Control:** Unlike the CG stage, the Cascode design allows for deterministic gain tuning. By setting the width ratio W_1/W_2 , we can precisely control

the transconductance ratio g_{m1}/g_{m2} , making the current transfer ratio highly predictable.

- **Design Efficiency:** The Cascode structure decouples the bias requirement from the buffering performance. Once M_2 is sized to establish the required DC operating point (V_S), M_1 can be sized independently to meet the current gain target, simplifying the design iteration process.
- **Improved Robustness:** While the gain is slightly lower due to the shunting effect of M_2 , the Cascode structure offers superior output impedance and a stabilized DC operating point for M_1 , leading to a more robust and scalable design.